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EXPERIMENT :-6

NAME:-Jashanpreet singh **UID:-25MCA20016**
BRANCH:-MCA **SECTION/GROUP:-6/B**
SEMESTER:-3rd **DATE:-21-10-2026**
SUBJECT:-AUTOMATION SOFTWARE TESTING **SUB.CODE:-25CAH-705**

AIM:-

a. Develop and execute automated test cases using Playwright to validate the search functionality of a web application by verifying relevant results for different input conditions.

b. Perform cross-browser testing using Playwright to ensure consistent behavior and functionality of the web application across multiple browsers.

c. Validate session management by automating login and logout operations, ensuring proper session creation, maintenance, and termination under various scenarios.

Web application used: Amazon India (https://www.amazon.in)

SOFTWARE/TOOLS REQUIRED:-

1. Operating System: macOS / Windows
2. Code Editor: Visual Studio Code (VS Code)
3. Programming Language: JavaScript
4. Automation Testing Tool: Playwright
5. Runtime Environment: Node.js and npm
6. Web Browsers: Google Chrome and Playwright WebKit
7. Websites Under Test:
   * Amazon India: https://www.amazon.in/
   * SauceDemo: https://www.saucedemo.com/
8. Terminal: VS Code Integrated Terminal
9. Assertion Library: Playwright Test (@playwright/test)
10. Test Execution Mode: Headed mode for visible browser execution

PROCEDURE:-

Part A: Amazon Search Functionality Testing

1. Open Visual Studio Code and create a Playwright project.
2. Install Node.js and the Playwright testing framework.
3. Create a test file named worksheet6.spec.js.
4. Open the Amazon India website using the browser.
5. Verify that the Amazon homepage and search bar are visible.

6. Enter “Samsung” in the search bar and click the search button.

7. Verify that the search results contain products.

8. Enter an invalid product name and verify that the search results page opens.

9. Submit an empty search and verify that the Amazon homepage remains accessible.

10. Execute the test cases using Playwright and check the results.

Part B: Amazon Cross-Browser Testing

1. Configure Playwright to run tests in Chrome and WebKit.

2. Open the Amazon India homepage in each browser.

3. Verify that the homepage and search bar are displayed correctly.

4. Search for “Samsung phone” and verify that the search results are displayed.

5. Search for an invalid product and verify that the website handles the query.

6. Click the account or sign-in option.

7. Verify that the Amazon sign-in page opens and the email field and Continue button are visible.

8. Compare the test results across both browser engines.

Part C: SauceDemo Session Management Testing

1. Open the SauceDemo website.

2. Enter the valid username standard_user and password secret_sauce.

3. Click the Login button and verify that the Products inventory page opens.

4. Refresh the inventory page and verify that the user remains logged in.

5. Open the navigation menu and click Logout.

6. Verify that the user is redirected to the login page.

7. After logout, attempt to access the inventory page directly and verify that access is denied.

8. Enter invalid login credentials and click Login.

9. Verify that an appropriate error message appears and the user cannot log in.

10. Execute the test cases using Playwright and review the pass/fail results in the terminal.

A. SEARCH FUNCTIONALITY TESTING:-

```
const { test, expect } = require("@playwright/test");

test("Amazon Search Functionality", async ({ page }) => {
   // Open Amazon India
   await page.goto("https://www.amazon.in/");

   // Search for Samsung products
   await page.locator("#twotabsearchtextbox").fill("Samsung");
   await page.locator("#nav-search-submit-button").click();
```

```
  // Verify search results
   await expect(page).toHaveURL(/k=Samsung/);
   await expect(
      page.locator('[data-component-type="s-search-result"]').first()
   ).toBeVisible();

   console.log("Search functionality test passed.");
});
```

OUTPUT:-

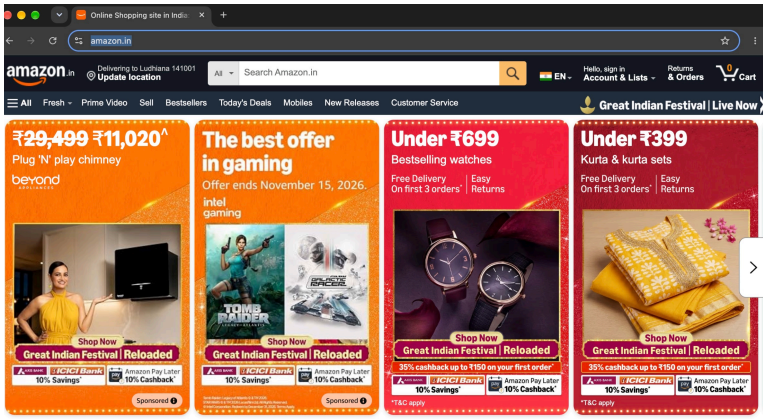
```
(base) jashanpreetsingh@Mac AmazonTesting % npx playwright test --project="Chrome" --grep "PART A" --headed

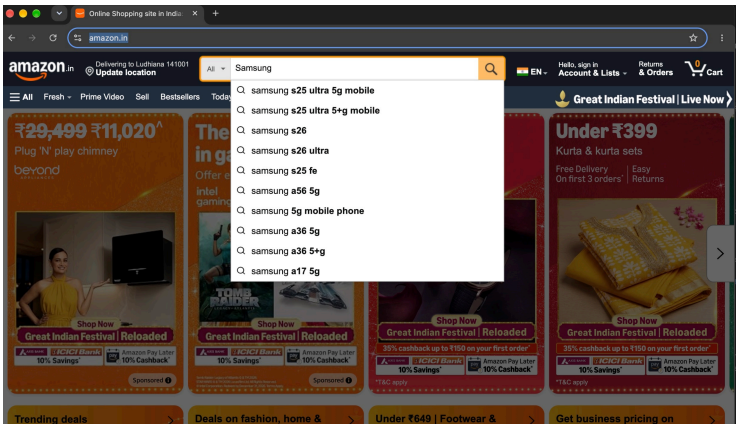
Running 3 tests using 1 worker

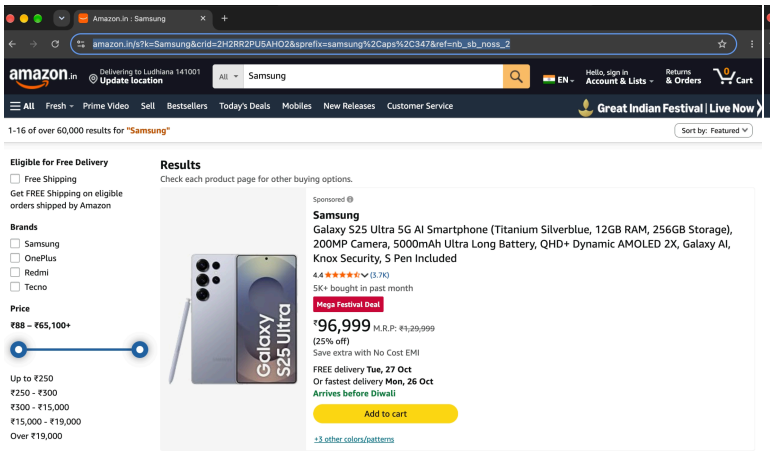
  ✓  1 [Chrome] › tests/worksheet6.spec.js:57:3 › PART A – Amazon India Search › TC01 – Search for Samsung products (4.9s)
TC01 PASSED: Samsung products displayed.
  ✓  2 [Chrome] › tests/worksheet6.spec.js:75:3 › PART A – Amazon India Search › TC02 – Search for an invalid product (4.3s)
TC02 PASSED: Invalid product search page displayed.
  ✓  3 [Chrome] › tests/worksheet6.spec.js:95:3 › PART A – Amazon India Search › TC03 – Submit an empty search (4.3s)
TC03 PASSED: Empty search handled.

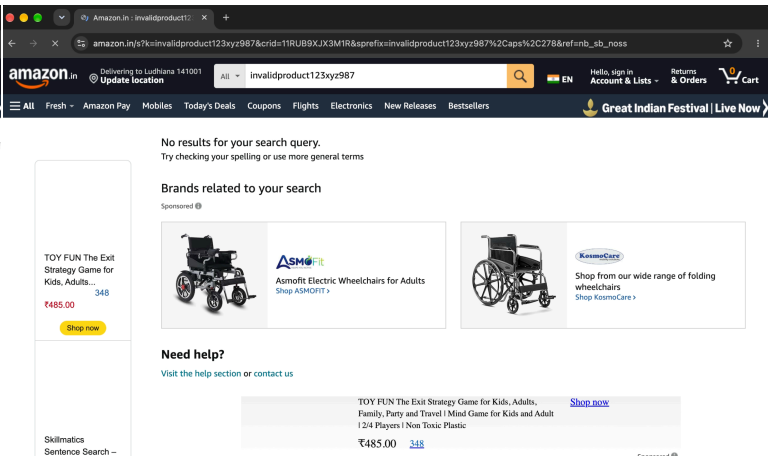
  3 passed (14.9s)
(base) jashanpreetsingh@Mac AmazonTesting %
```

Terminal output of the search tests









TEST CASES:-

Test Case ID	Test Case Description	Test Input	Expected Result	Actual Result	Status
TC_A01	Search box displayed	https://www.amazon.in	Home page opens; search box is visible and search button is enabled.	Search box was visible and button enabled.	Pass
TC_A02	Valid keyword	Samsung	Products related to the keyword are listed.	Relevant products were listed.	Pass
TC_A03	Multi word keyword	wireless bluetooth headphones	A list of products is displayed.	Product list was displayed.	Pass
TC_A04	Case insensitive search	MILTON WATER BOTTLE	Same kind of products are displayed.	Same kind of products displayed.	Pass
TC_A05	Submit an empty search	Empty input	The website should remain functional and handle the empty search.	Amazon homepage remains accessible.	Pass
TC_A06	Unknown keyword	invalidproduct123xyz987	No results message is displayed.	No results message displayed.	Pass

2. CROSS-BROWSER TESTING:-

CODING:- tests/amazon-crossbrowser.spec.js (runs every test on four browsers)
Setup and helper code (launches Chromium, Google Chrome, Safari (WebKit) and Firefox)

```
const { test, expect } = require("@playwright/test");
test("Amazon Cross-Browser Testing", async ({ page }) => {
    // Open Amazon India
    await page.goto("https://www.amazon.in/");
    // Verify homepage
    await expect(page).toHaveTitle(/Amazon/i);
    // Search for Samsung phones
    await page.locator("#twotabsearchtextbox").fill("Samsung phone");
    await page.locator("#nav-search-submit-button").click();
    // Verify search results
    await expect(page).toHaveURL(/k=Samsung/);
    await expect(
        page.locator('[data-component-type="s-search-result"]').first()
    ).toBeVisible();
    console.log("Cross-browser test passed.");
});
```

OUTPUT:-

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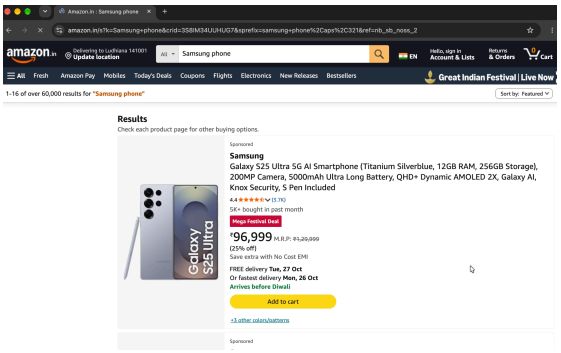
```
(base) jashanpreetsingh@Mac AmazonTesting % npx playwright test --grep "PART B" --workers=1

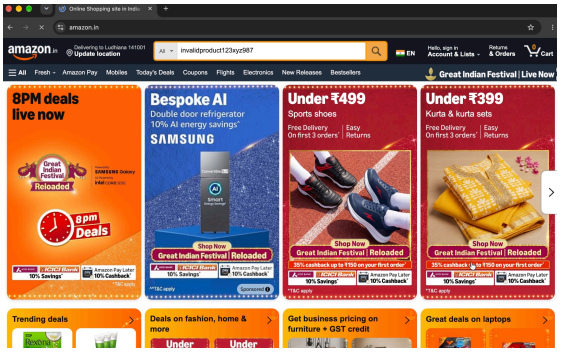
Running 8 tests using 1 worker

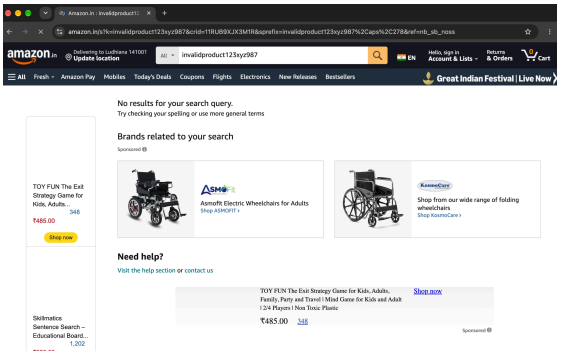
  ✓  1 … B – Amazon Cross-Browser Testing › TC_B01 – Amazon home page and search are available (2.7s)
  ✓  2 …67:3 › PART B – Amazon Cross-Browser Testing › TC_B02 – Search shows relevant products (4.3s)
  ✓  3 …c.js:77:3 › PART B – Amazon Cross-Browser Testing › TC_B03 – Invalid search is handled (4.3s)
  ✓  4 …et6.spec.js:85:3 › PART B – Amazon Cross-Browser Testing › TC_B04 – Sign-in page opens (4.1s)
  ✓  5 … B – Amazon Cross-Browser Testing › TC_B01 – Amazon home page and search are available (3.2s)
  ✓  6 …67:3 › PART B – Amazon Cross-Browser Testing › TC_B02 – Search shows relevant products (4.2s)
  ✓  7 …c.js:77:3 › PART B – Amazon Cross-Browser Testing › TC_B03 – Invalid search is handled (4.2s)
  ✓  8 …et6.spec.js:85:3 › PART B – Amazon Cross-Browser Testing › TC_B04 – Sign-in page opens (3.7s)

  8 passed (32.8s)
(base) jashanpreetsingh@Mac AmazonTesting %
```

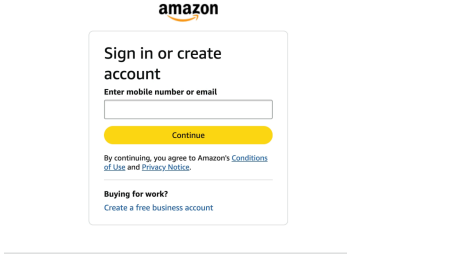
Chrome



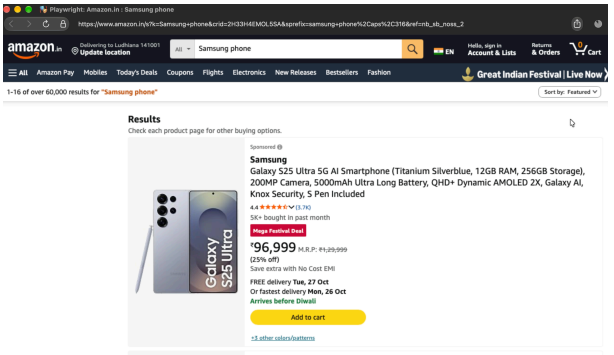


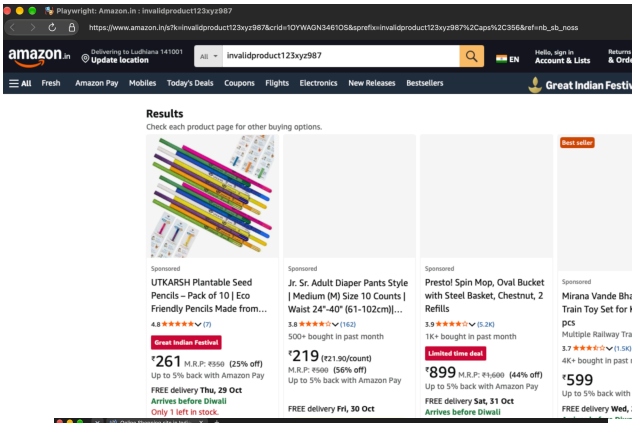


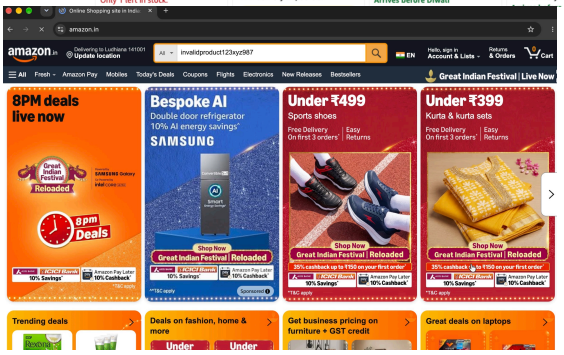




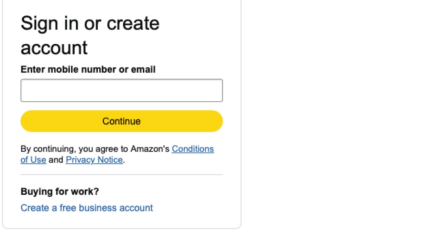
safari











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TEST CASES:-

Test Case ID	Test Case Description	Test Input	Expected Result	Actual Result	Status
TC_B01	Home page loads	https://www.amazon.in	Title, logo and search box are visible in all browsers.	Home page loaded in all browsers.	Pass
TC_B02	Search shows products	Search for Samsung phones	Products are listed in all browsers.	Products listed in all browsers.	Pass
TC_B03	Search for an invalid product	invalidproduct123xyz987	Website should handle the invalid search without crashing.	Same behavior in all browsers.	Pass
TC_B04	Sign in page	https://www.amazon.in	Sign in page with the email field opens in all browsers.	Sign in page opened in all browsers.	Pass

2.C. SESSION MANAGEMENT TESTING:-

CODING:- (sign in is done once in the browser window)

```
const { test, expect } = require("@playwright/test");
test("SauceDemo Session Management", async ({ page }) => {
    // Open SauceDemo
    await page.goto("https://www.saucedemo.com/");
    // Login with valid credentials
    await page.locator("#user-name").fill("standard_user");
    await page.locator("#password").fill("secret_sauce");
    await page.locator("#login-button").click();
    // Verify successful login
    await expect(page).toHaveURL(/inventory\.html/);
    // Refresh and verify session
    await page.reload();
    await expect(page.locator(".title")).toHaveText("Products");
    // Logout
    await page.locator("#react-burger-menu-btn").click();
    await page.locator("#logout_sidebar_link").click();
    // Verify logout
    await expect(page.locator("#login-button")).toBeVisible();
    console.log("Session management test passed.");
});
```

OUTPUT:-

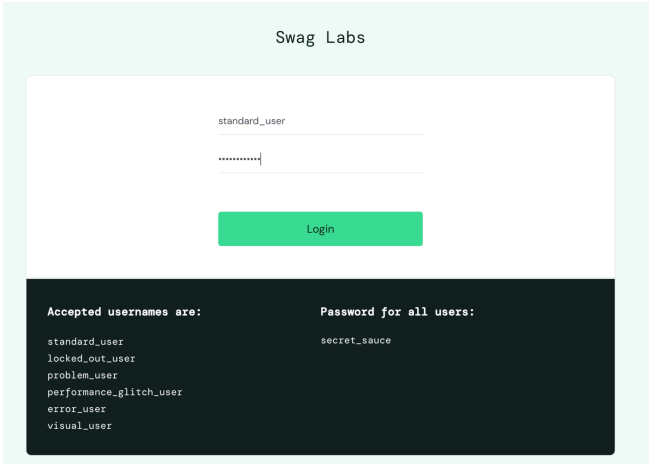
```
(base) jashanpreetsingh@Mac AmazonTesting % npx playwright test --project=chrome --grep "PART C" --headed

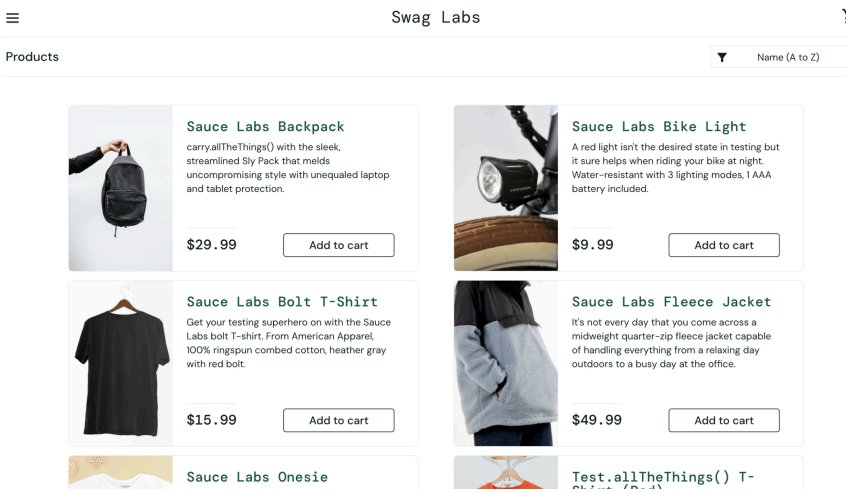
Running 5 tests using 1 worker

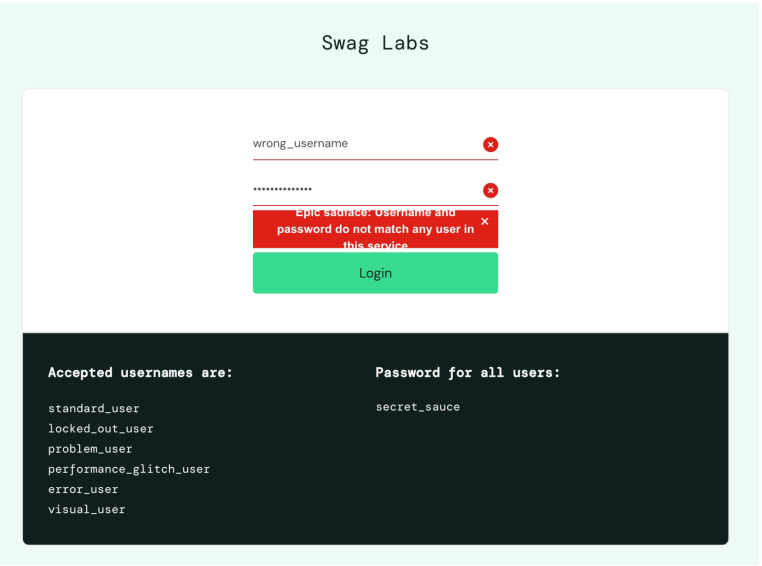
  ✓  1 [Chrome] › tests/worksheet6.spec.js:223:3 › PART C – SauceDemo Session Management › TC01 – Valid login (4.1s)
TC01 PASSED: User logged in and inventory opened.
  ✓  2 …rome] › tests/worksheet6.spec.js:237:3 › PART C – SauceDemo Session Management › TC02 – Session maintenance after refresh (4.9s)
TC02 PASSED: Session maintained after refresh.
  ✓  3 [Chrome] › tests/worksheet6.spec.js:255:3 › PART C – SauceDemo Session Management › TC03 – Logout (5.4s)
TC03 PASSED: User logged out successfully.
  ✓  4 [Chrome] › tests/worksheet6.spec.js:279:3 › PART C – SauceDemo Session Management › TC04 – Access after logout (6.5s)
TC04 PASSED: Inventory access denied after logout.
  ✓  5 [Chrome] › tests/worksheet6.spec.js:303:3 › PART C – SauceDemo Session Management › TC05 – Invalid login (3.9s)
TC05 PASSED: Invalid login rejected.

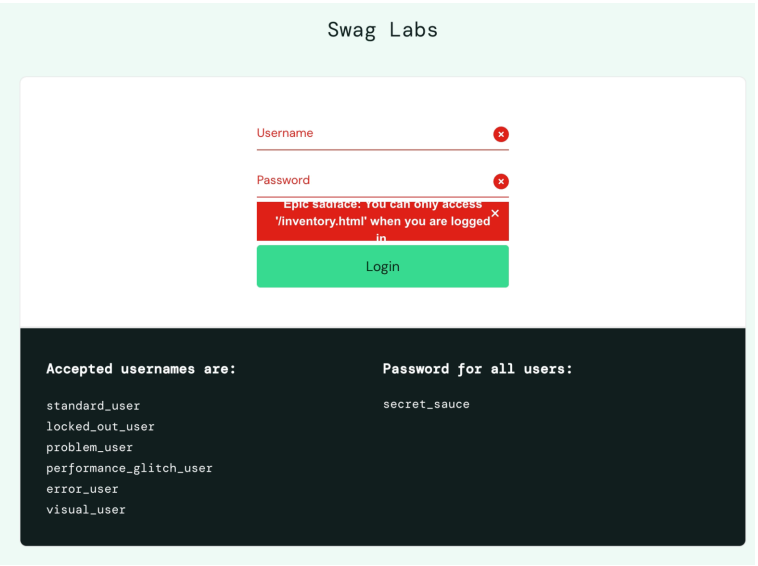
  5 passed (25.7s)
(base) jashanpreetsingh@Mac AmazonTesting %
```

Terminal output of the session tests (3 passed)









TEST CASES:-

Test Case ID	Test Case Description	Test Input	Expected Result	Actual Result	Status
TC_C01	Login creates session	Username: standard_user, Password: secret_sauce	User should log in and access the inventory page.	Inventory page opened successfully.	Pass
TC_C02	Session after reload	Refresh the inventory page	User stays logged in.	User stayed logged in.	Pass
TC_C03	Verify logout functionality	Click the Logout button	User should be redirected to the login page.	User should be redirected to the login page.	Pass
TC_C04	Verify access after logout	Open the inventory URL after logout	User should be redirected to the login page.	Access denied after logout.	Pass
TC_C05	Verify invalid login	Username: wrong_username, Password: wrong_password	An error message should appear, and login should fail.	Error message displayed successfully.	Pass

RESULT:-

Successfully performed Search Functionality Testing, Cross-Browser Testing, and Session Management Testing using Playwright with JavaScript. Amazon India search operations were tested, website functionality was checked across Chrome and WebKit, and SauceDemo login, session maintenance, logout, and invalid login scenarios were verified. The tests helped evaluate the functionality and reliability of the websites under different testing conditions.

LEARNING OUTCOMES WHAT I HAVE LEARNT:-

1. Learnt to automate search testing on a real e-commerce website using different input conditions.
2. Learnt to run the same Playwright tests on Chromium, Google Chrome, Safari (WebKit) and Firefox.
3. Learnt to automate sign in and sign out and verify session creation, maintenance and termination.
4. Learnt to test login, session and logout without storing a password in the code, and to read the Playwright report.

EVALUATION GRID:-

SR.NO	PARAMETERS	MARKS OBTAINED	MAXIMUM MARKS
1	**Worksheet Completion**		**08 marks**
2	**Conduct**		**12 marks**
3	**Viva-Voice**		**10 marks**
	Total		**30 marks**